

Assessment 1 - COVID-19 impact on digital learning

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1. Data Cleaning and Wrangling

Describe and show the data cleaning, transformation and wrangling process' that you have completed. Ensure you provide justification of any data handling and transformation you have completed.

The first step is to read in and clean the districts information data. A comprehensive analysis was done to identify any missing values and clean the dataset.

```
# Checking for missing values in the districts data  
anyNA(district_data)
```

```
## [1] FALSE
```

```
sum(is.na(district_data))
```

```
## [1] 0
```

```
colSums(is.na(district_data))
```

```
##           district_id           state           locale  
##                0                0                0  
##    pct_black.hispanic    pct_free.reduced county_connections_ratio  
##                0                0                0  
##           pp_total_raw  
##                0
```

```
# Removing duplicate rows  
district_data <- district_data %>% distinct()
```

- We could see some inconsistencies in the dataset.
- I have done the following to rectify these errors:

```
district_data <- district_data %>%  
  mutate(state = case_when(  
    state == "Utaah" ~ "Utah",  
    state == "UTAH" ~ "Utah",  
    state == "uTthah" ~ "Utah",  
    state == "New YOrk" ~ "New York",
```

```

    state == "NY City" ~ "New York",
    state == "Califronia" ~ "California",
    state == "ConnectiCUT" ~ "Connecticut",
    TRUE ~ state
  ))
district_data <- district_data %>%
mutate(locale = case_when(
  locale == "Sub" ~ "Suburb",
  locale == "Cit" ~ "City",
  locale == "CItty" ~ "City",
  TRUE ~ locale
))

```

I have done the same cleaning process with the products info dataset as well

```
anyNA(product_data)
```

```
## [1] FALSE
```

```
sum(is.na(product_data))
```

```
## [1] 0
```

```
colSums(is.na(product_data))
```

```
##      lp_id      url product_name      company      sector      pef
##         0         0             0           0           0         0
##    pef_cat  pef_subcat
##         0           0
```

```

# Removing duplicate rows
product_data <- product_data %>% distinct()

```

```
unique(product_data$LP.ID)
```

```
## NULL
```

```
unique(product_data$URL)
```

```
## NULL
```

```
unique(product_data$Product.Name)
```

```
## NULL
```

```
unique(product_data$Provider.Company.Name)
```

```
## NULL
```

```
unique(product_data$Sector.s.)
```

```
## NULL
```

```
unique(product_data$Primary.Essential.Function)
```

```
## NULL
```

- We could see some inconsistencies in the dataset.
- I have done the following to rectify these errors:

```
product_data=product_data %>%  
  mutate(company = replace(company,company == "Adobe Inc","Adobe Inc. ")) %>%  
  mutate(sector = replace(sector,sector == "PreK-112","PreK-12")) %>%  
  mutate(sector = replace(sector,sector == "PPreK-12","PreK-12")) %>%  
  mutate(sector = replace(sector,sector == "PreK-122","PreK-12")) %>%  
  mutate(sector = replace(sector,sector == "PreK-12; Higher; Corporate","PreK-12; Higher Ed; Corporate").
```

The next step is to clean the engagement data to whatever extent possible, keeping in mind the limitations mentioned in the ReadMe file.

There were significant inconsistencies in the engagement data, specifically in the date columns which had to be rectified.

```
# Extracting the year and changing it's format to numeric  
district_1000$year <- as.numeric(str_sub(district_1000$time,-4))  
unique(district_1000$year)
```

```
## [1] 2020 2022 1020
```

```
district_1039$year <- as.numeric(str_sub(district_1039$time,-4))  
unique(district_1039$year)
```

```
## [1] 2020 2044
```

```
district_1044$year <- as.numeric(str_sub(district_1044$time,-4))  
unique(district_1044$year)
```

```
## [1] 2020 2050
```

```
district_1052$year <- as.numeric(str_sub(district_1052$time,-4))  
unique(district_1052$year)
```

```
## [1] 2020 2033
```

```
district_1131$year <- as.numeric(str_sub(district_1131$time,-4))  
unique(district_1131$year)
```

```
## [1] 2020 2050
```

```
#extract the month and convert to numeric format
district_1000$month <- as.character(str_sub(district_1000$time,-7,-6))
district_1039$month <- as.character(str_sub(district_1039$time,-7,-6))
district_1044$month <- as.character(str_sub(district_1044$time,-7,-6))
district_1052$month <- as.character(str_sub(district_1052$time,-7,-6))
district_1131$month <- as.character(str_sub(district_1131$time,-7,-6))
```

This discrepancy led to the conclusion that the data points were probably entered incorrectly. In order to guarantee data consistency and conformity to the designated data year, I decided to standardize all of these data points to the year 2020 in order to address this problem.

```
#change the date type
district_1000_2 <- d1000_clean %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
  str(district_1000_2)
```

```
## 'data.frame': 104003 obs. of 7 variables:
## $ time : Date, format: "2020-01-01" "2020-01-01" ...
## $ lp_id : int 93690 17941 65358 98265 59257 90153 41587 29322 37479 51340 ...
## $ pct_access : chr "0" "0.03" "0.03" "0.57" ...
## $ engagement_index: chr "" "0.9" "1.2" "37.79" ...
## $ district_id : num 1000 1000 1000 1000 1000 1000 1000 1000 1000 1000 ...
## $ year : chr "2020" "2020" "2020" "2020" ...
## $ month : chr "01" "01" "01" "01" ...
```

```
district_1039_2 <- district_1039 %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
  str(district_1039_2)
```

```
## 'data.frame': 38791 obs. of 7 variables:
## $ time : Date, format: "2020-01-01" "2020-01-01" ...
## $ lp_id : chr "64947" "52758" "25559" "13496" ...
## $ pct_access : num 0.09 0.09 0.09 0.09 0.09 0.91 0.27 0.18 0.09 0.09 ...
## $ engagement_index: num 0.91 1.83 13.71 4.57 1.83 ...
## $ district_id : num 1039 1039 1039 1039 1039 ...
## $ year : num 2020 2020 2020 2020 2020 ...
## $ month : chr "/1" "/1" "/1" "/1" ...
```

```
district_1044_2 <- district_1044 %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
  str(district_1044_2)
```

```
## 'data.frame': 255722 obs. of 7 variables:
## $ time : Date, format: "2020-01-01" "2020-01-01" ...
## $ lp_id : int 99792 80493 16164 61384 88925 30114 57293 85603 83248 52409 ...
## $ pct_access : chr "0.02" "0.01" "0.03" "0.01" ...
## $ engagement_index: chr "2.26" "0.17" "0.35" "16.52" ...
## $ district_id : num 1044 1044 1044 1044 1044 ...
## $ year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ month : chr "01" "01" "01" "01" ...
```



```
district_1052_2 <- district_1052 %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
  str(district_1052_2)
```

```
## 'data.frame':    91977 obs. of  7 variables:
## $ time          : Date, format: "2020-01-01" "2020-01-01" ...
## $ lp_id         : int  57513 42091 53399 32213 85836 94158 26488 99916 92867 16164 ...
## $ pct_access    : num  0.49 0.04 0.04 0.9 0.04 0.04 0.41 1.19 0.04 0.04 ...
## $ engagement_index: num  141.28 0.41 8.19 44.23 0.41 ...
## $ district_id   : num  1052 1052 1052 1052 1052 ...
## $ year          : num  2020 2020 2020 2020 2020 ...
## $ month         : chr  "01" "01" "01" "01" ...
```

```
district_1131_2 <- district_1131 %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
  str(district_1131_2)
```

```
## 'data.frame':    32963 obs. of  7 variables:
## $ time          : Date, format: "2020-01-01" "2020-01-01" ...
## $ lp_id         : int  16753 80560 85682 95731 28504 57084 25298 32213 50479 96111 ...
## $ pct_access    : num  0.04 0.09 0 0.09 0 0.04 0.04 0.04 0.04 0 ...
## $ engagement_index: num  3.58 8.94 NA 2.24 NA 0.89 0.45 0.45 0.45 NA ...
## $ district_id   : num  1131 1131 1131 1131 1131 ...
## $ year          : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ month         : chr  "01" "01" "01" "01" ...
```

```
# extract the year and convert to numeric format
district_1000_2$year <- as.numeric(format(district_1000_2$time, "%Y"))
unique(district_1000_2$year)
```

```
## [1] 2020
```

```
district_1039_2$year <- as.numeric(format(district_1039_2$time, "%Y"))
unique(district_1039_2$year)
```

```
## [1] 2020 NA
```

```
district_1044_2$year <- as.numeric(format(district_1044_2$time, "%Y"))
unique(district_1044_2$year)
```

```
## [1] 2020
```

```
district_1052_2$year <- as.numeric(format(district_1052_2$time, "%Y"))
unique(district_1052_2$year)
```

```
## [1] 2020
```

```
district_1131_2$year <- as.numeric(format(district_1131_2$time, "%Y"))
unique(district_1131_2$year)
```

```
## [1] 2020
```

```
district_1000_2 <- district_1000 %>%
  filter(year != 2010)
```

The product information data set contains some duplicates, as was previously indicated. Therefore, in order to improve the analysis's correctness, the lp_id in each district file needs to be replaced.

-The goal of removing NA values from each district's file is to make the dataset more streamlined by removing unnecessary rows. To determine whether the associated 'engagement_index' is also NA when the 'pct_access' column includes NA, a manual analysis was carried out. When the values in both columns are zero, the data is considered unnecessary and should be removed. This proactive approach to data management guarantees the relevance and utility of the dataset while also improving memory and space efficiency.

```
#Removing any missing values from district 1000
anyNA(d1000_clean)
```

```
## [1] TRUE
```

```
sum(is.na(d1000_clean))
```

```
## [1] 2
```

```
colSums(is.na(d1000_clean))
```

```
##           time           lp_id      pct_access engagement_index
##           0             2           0              0
##  district_id         year      month
##           0             0           0
```

```
d1000_Final <- na.omit(d1000_clean)
```

```
#Removing any missing values from district 1039
anyNA(d1039_clean)
```

```
## [1] FALSE
```

```
sum(is.na(d1039_clean))
```

```
## [1] 0
```

```
colSums(is.na(d1039_clean))
```

```
##           time           lp_id      pct_access engagement_index
##           0             0           0              0
##  district_id         year      month
##           0             0           0
```

```
d1039_Final <- na.omit(d1039_clean)
```

```
#Removing any missing values from district 1044  
anyNA(d1044_clean)
```

```
## [1] FALSE
```

```
sum(is.na(d1044_clean))
```

```
## [1] 0
```

```
colSums(is.na(d1044_clean))
```

```
##           time           lp_id      pct_access engagement_index  
##           0             0           0             0  
##  district_id         year         month  
##           0             0           0
```

```
d1044_Final <- na.omit(d1044_clean)
```

```
#Removing any missing values from district 1052  
anyNA(d1052_clean)
```

```
## [1] TRUE
```

```
sum(is.na(d1052_clean))
```

```
## [1] 30762
```

```
colSums(is.na(d1052_clean))
```

```
##           time           lp_id      pct_access engagement_index  
##           0             0           0             30762  
##  district_id         year         month  
##           0             0           0
```

```
d1052_Final <- na.omit(d1052_clean)
```

```
#Removing any missing values from district 1131  
anyNA(d1131_clean)
```

```
## [1] TRUE
```

```
sum(is.na(d1131_clean))
```

```
## [1] 16554
```

```
colSums(is.na(d1131_clean))
```

```
##           time           lp_id      pct_access engagement_index
##           0             0           0             16554
##   district_id         year         month
##           0             0             0
```

```
d1131_Final <- na.omit(d1131_clean)
```

- The next step is to combine all of the 5 district engagement datasets to create 1 main dataset, which makes it easier for data analysis.

```
# Combining the engagement datasets into one dataframe
```

```
district_final <- bind_rows(d1000_Final, d1039_Final, d1044_Final, d1052_Final, d1131_Final)
district_final2 <- bind_rows(d1000_Final, d1039_Final, d1044_Final, d1052_Final, d1131_Final)
district_final4 <- bind_rows(d1000_Final, d1039_Final, d1044_Final, d1052_Final, d1131_Final)
```

```
# Convert `lp_id` to character in both data frames
```

```
district_final4$lp_id <- as.character(district_final4$lp_id)
product_data$lp_id <- as.character(product_data$lp_id)
```

```
# Convert `lp_id` to character in both data frames
```

```
district_final2$lp_id <- as.character(district_final2$lp_id)
product_data$lp_id <- as.character(product_data$lp_id)
```

```
# Convert the 'time' column in the district_final dataframe to a Date type
```

```
district_final <- district_final %>%
  mutate(time = as.Date(time, "%d/%m/%y"))
district_final$district_id =
  as.character(district_final$district_id)
district_final$lp_id =
  as.character(district_final$lp_id)
```

```
# Convert `lp_id` to character in both data frames
```

```
district_final$lp_id <- as.character(district_final$lp_id)
product_data$lp_id <- as.character(product_data$lp_id)
```

```
# Combining district combination in the context of the district and product info
```

```
top372 <- left_join(district_final, product_data, by = c("lp_id" = "lp_id"))
top372 <- top372 %>%
  select(-time) %>%
  mutate_all(~ ifelse(. == "", "Unknown", .)) %>%
  mutate_all(~ ifelse(is.na(.), "Unknown", .))
```

```
top372 <- top372 %>%
  mutate(engagement_index = as.numeric(engagement_index))
```

```

product_rank_1 <- top372 %>%
  group_by(lp_id) %>%
  summarise(sum_engagement = sum(engagement_index, na.rm = TRUE), .groups = "drop")

product_rank_2 <- product_rank_1 %>%
  arrange(desc(sum_engagement)) %>%
  mutate(rank = row_number(desc(sum_engagement)))

product_rank_3 <- left_join(top372, product_rank_2 %>% select(lp_id, rank), by = "lp_id")

top_10_product <- product_rank_3 %>%
  filter(rank <= 10) %>%
  group_by(product_name, lp_id, month, .groups = "drop") %>%
  summarize(total_engagement = sum(engagement_index))

## 'summarise()' has grouped output by 'product_name', 'lp_id', 'month'. You can
## override using the '.groups' argument.

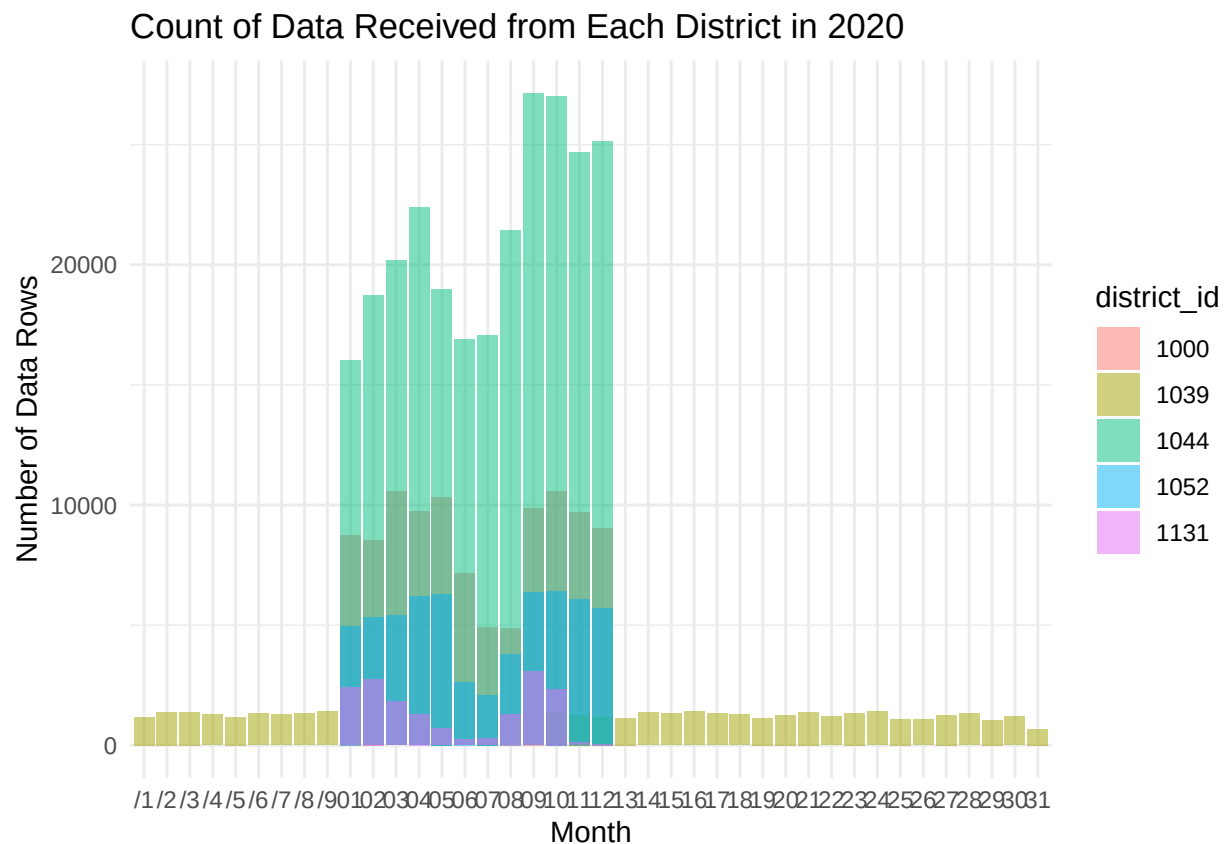
```

2. Data Visualisation

Provide evidence of your visualisation techniques and the reason for your choice of visual tools. Provide a description of each visualisation you are presenting. There is no need to show your code.

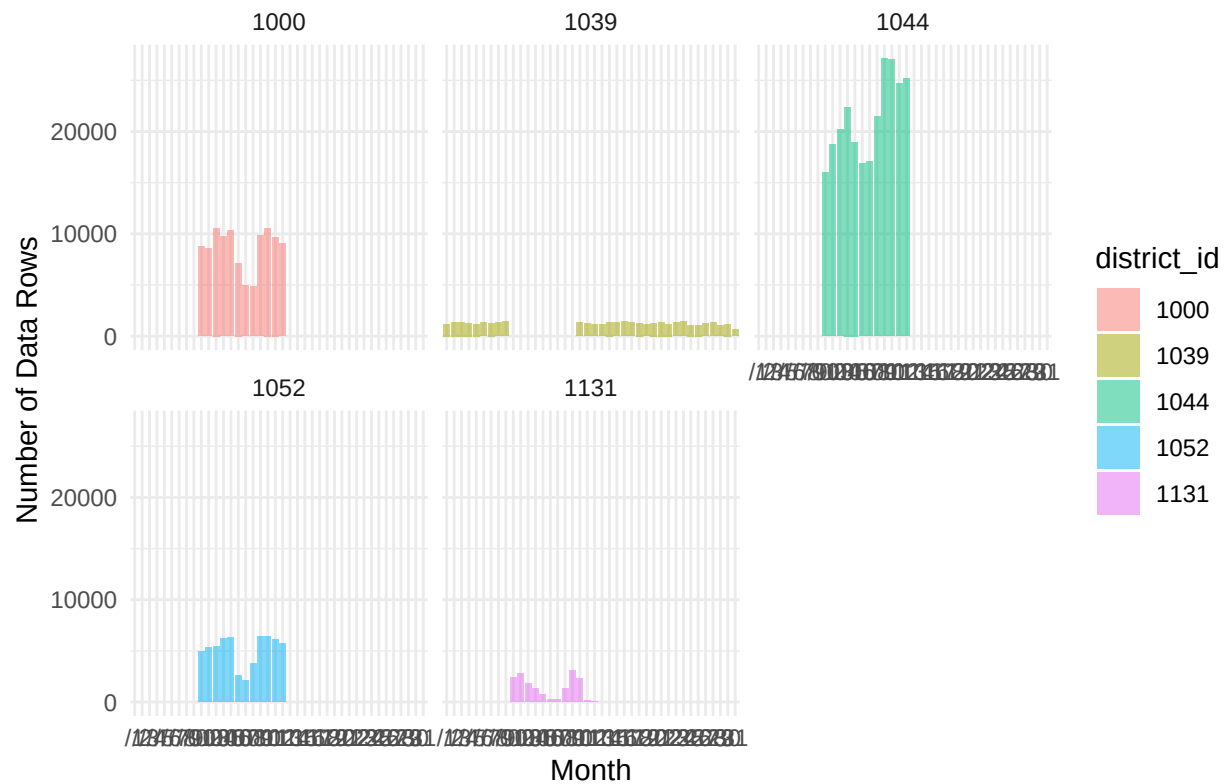
First let's look at the amount of engagement data we have from each district

```
ggplot(data=district_final4, aes(x=month, fill=district_id)) +  
  geom_bar(position="identity", alpha=0.5) +  
  theme_minimal() +  
  ggtitle("Count of Data Received from Each District in 2020") +  
  labs(x = "Month", y = "Number of Data Rows")
```



```
ggplot(data=district_final4, aes(x=month, fill = district_id)) +  
  geom_histogram (stat="count", binwidth=0.2, alpha=0.5) +  
  facet_wrap(~ district_id) +  
  theme_minimal() +  
  ggtitle("Count of Data Received from Each District in 2020") +  
  labs(x = "Month", y = "Number of Data Rows")
```

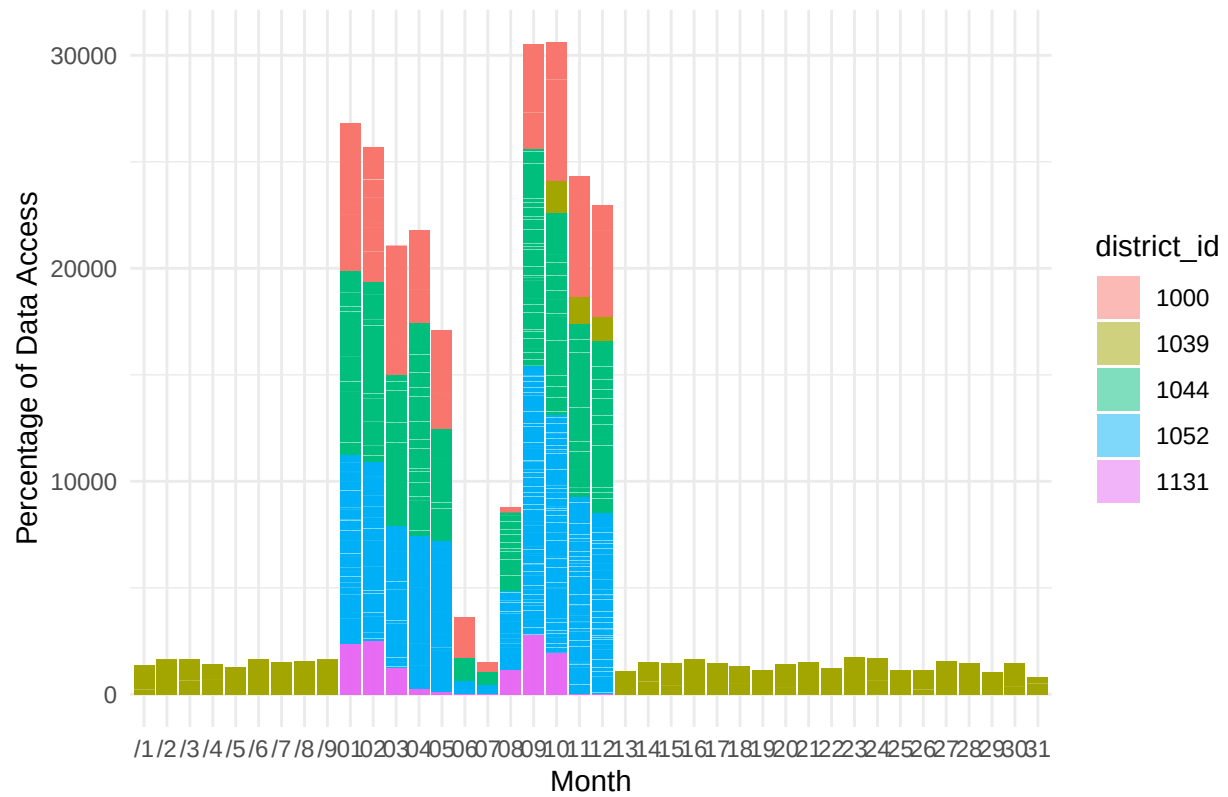
Count of Data Received from Each District in 2020



- Next, let's look at how the level of broadband access varies from each district

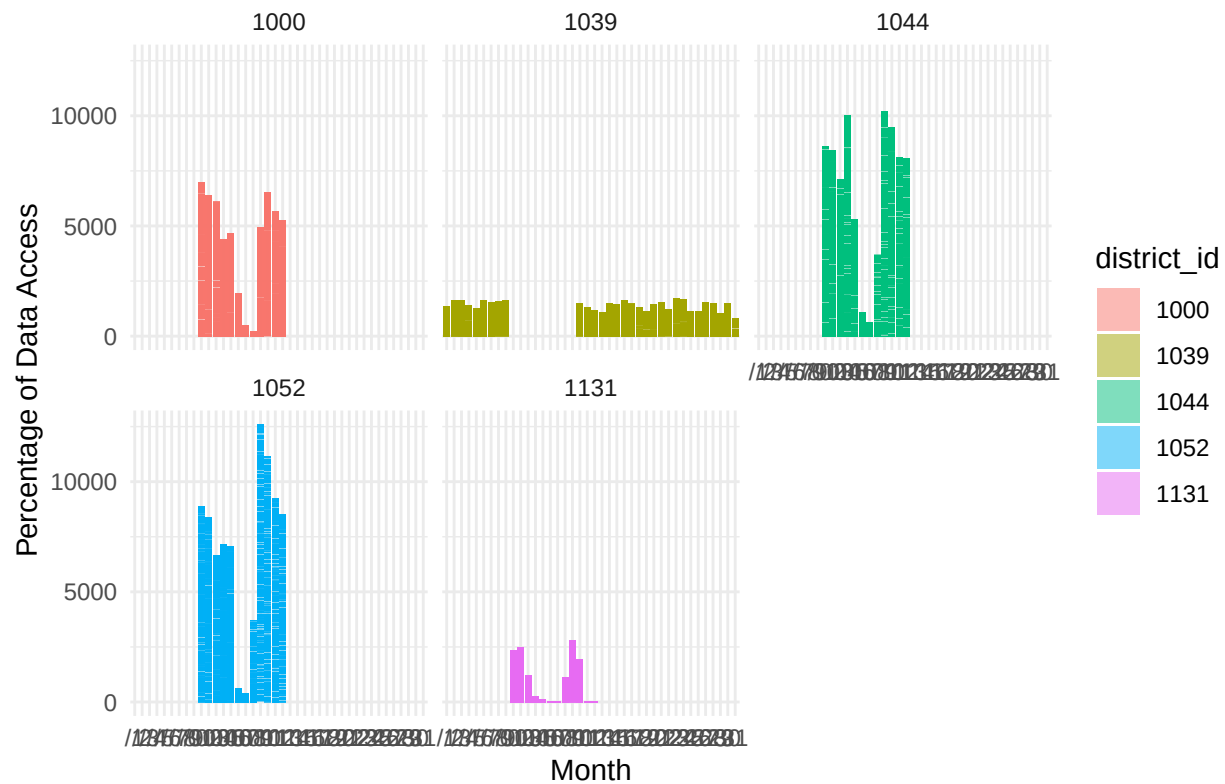
```
# geom_bar(stat = "identity", position = "stack", alpha = 0.5) +
# theme_minimal() +
# ggtitle("Percentage of Data Access from Each District in 2020")+
# labs(x = "Month", y = "Percentage of Data Access")
ggplot(data = district_final4, aes(x = month, y = pct_access, fill = district_id)) +
  geom_bar(stat = "identity", position = "stack", alpha = 0.5) +
  theme_minimal() +
  ggtitle("Percentage of Data Access from Each District in 2020") +
  labs(x = "Month", y = "Percentage of Data Access")
```

Percentage of Data Access from Each District in 2020



```
ggplot(data = district_final4, aes(x = month, y = pct_access, fill = district_id)) +
  geom_bar(stat = "identity", position = "stack", alpha = 0.5) +
  theme_minimal() +
  facet_wrap(~ district_id) +
  ggtitle("Percentage of Data Access from Each District in 2020") +
  labs(x = "Month", y = "Percentage of Data Access")
```


Percentage of Data Access from Each District in 2020

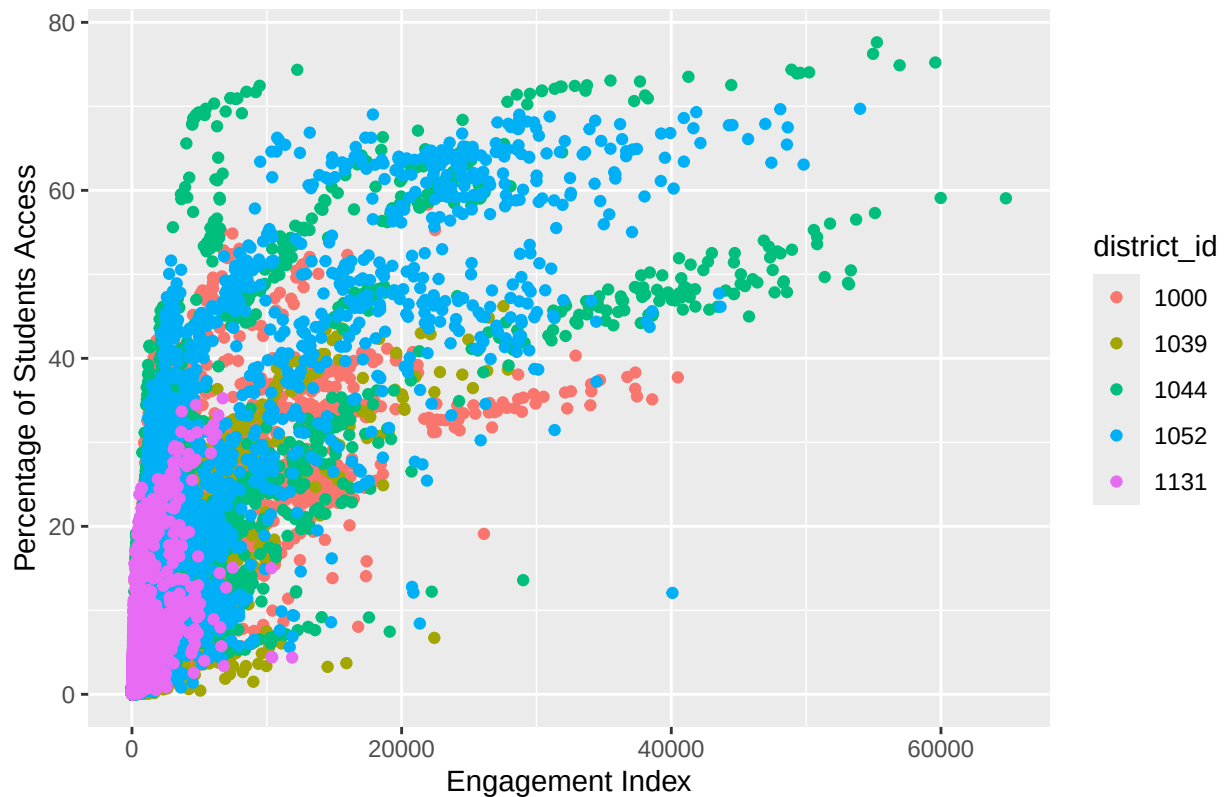


1. What is the picture of digital connectivity and engagement in 2020?

- As a first step towards analyzing the data, a preliminary scatterplot can be used to visualize the relationship between engagement and broadband access(connectivity). This helps in identifying trends in the data which can be further explored.

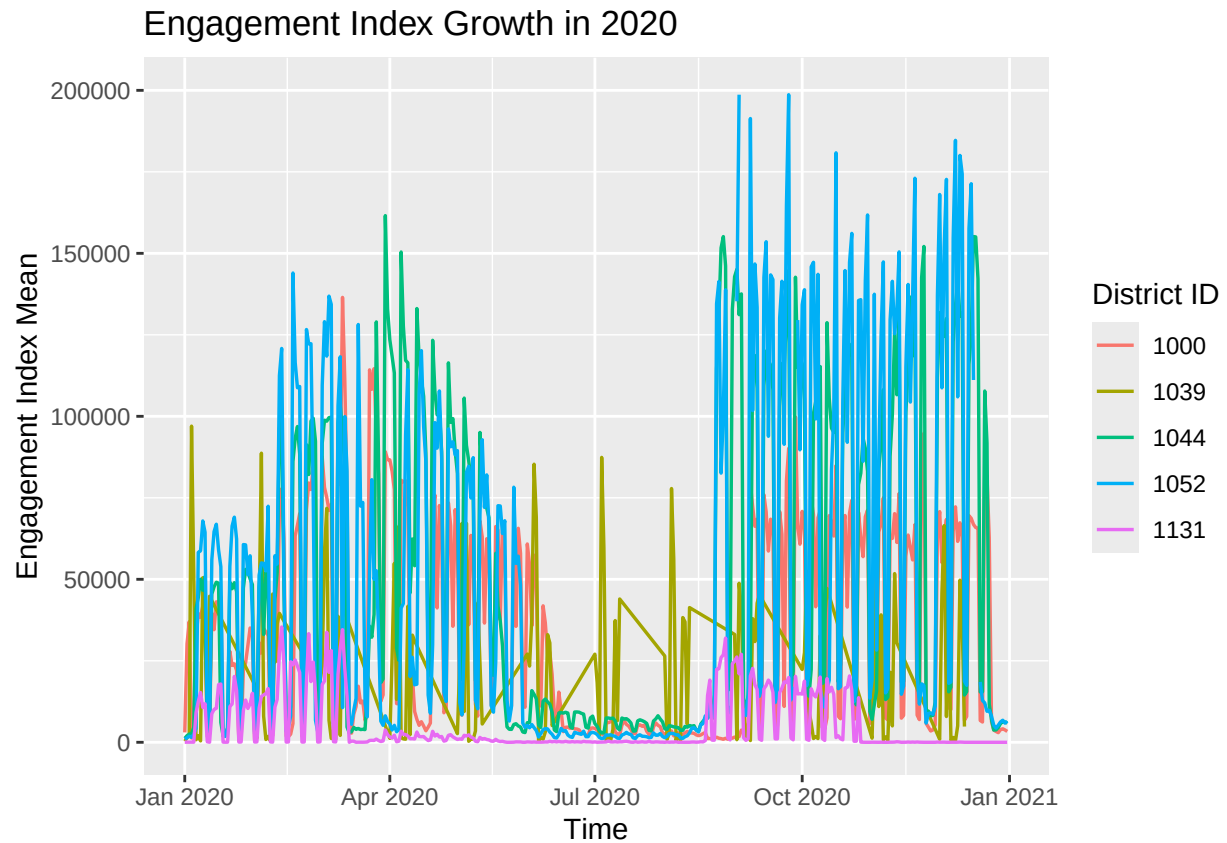
```
# ggplot(district_final) +
#   geom_point(aes(x=engagement_index, y=pct_access, color = district_id))+
#   xlab("Engagement Index")+
#   ylab("Percentage of Students Access")+
#   ggtitle("Correlation of broadband access and engagement in 2020")
ggplot(data = district_final, aes(x = engagement_index, y = pct_access, color = district_id)) +
  geom_point() +
  labs(x = "Engagement Index", y = "Percentage of Students Access", title = "Correlation of broadband a
```

Correlation of broadband access and engagement in 2020



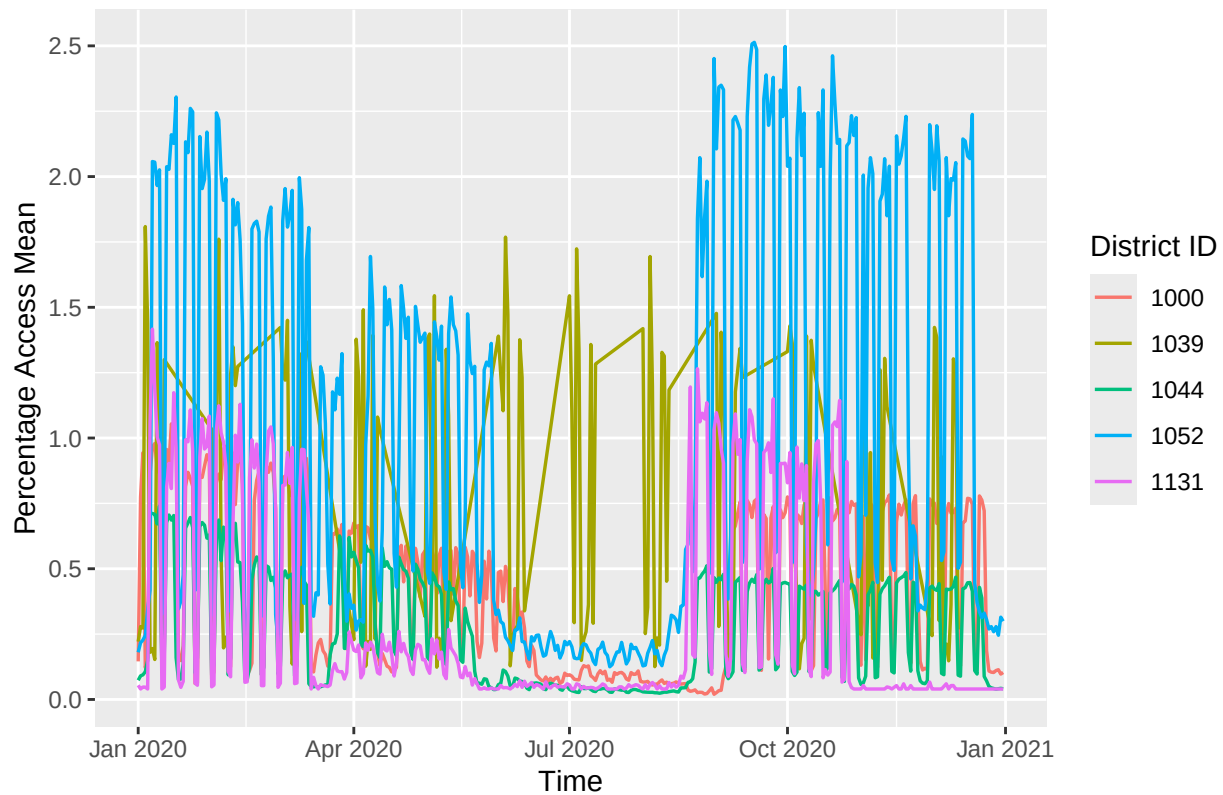
2. What is the effect of the COVID-19 pandemic on online and distance learning, and how might this also evolve in the future?

```
ggplot(data1, aes(x = time, y = sum_engagement, color = district_id)) +
  geom_line(linewidth=0.6) +
  xlab("Time") +
  ylab("Engagement Index Mean") +
  ggtitle("Engagement Index Growth in 2020")+
  xlim(as.Date("2020-01-01"), as.Date("2020-12-31"))+
  ylim(0, 200000) +
  scale_color_discrete(name = "District ID")
```



```
ggplot(data2, aes(x = time, y = mean_pctaccess, color = district_id)) +
  geom_line(linewidth=0.6) +
  xlab("Time") +
  ylab("Percentage Access Mean") +
  ggtitle("Percentage Access Growth in 2020")+
  xlim(as.Date("2020-01-01"), as.Date("2020-12-31"))+
  scale_color_discrete(name = "District ID")
```

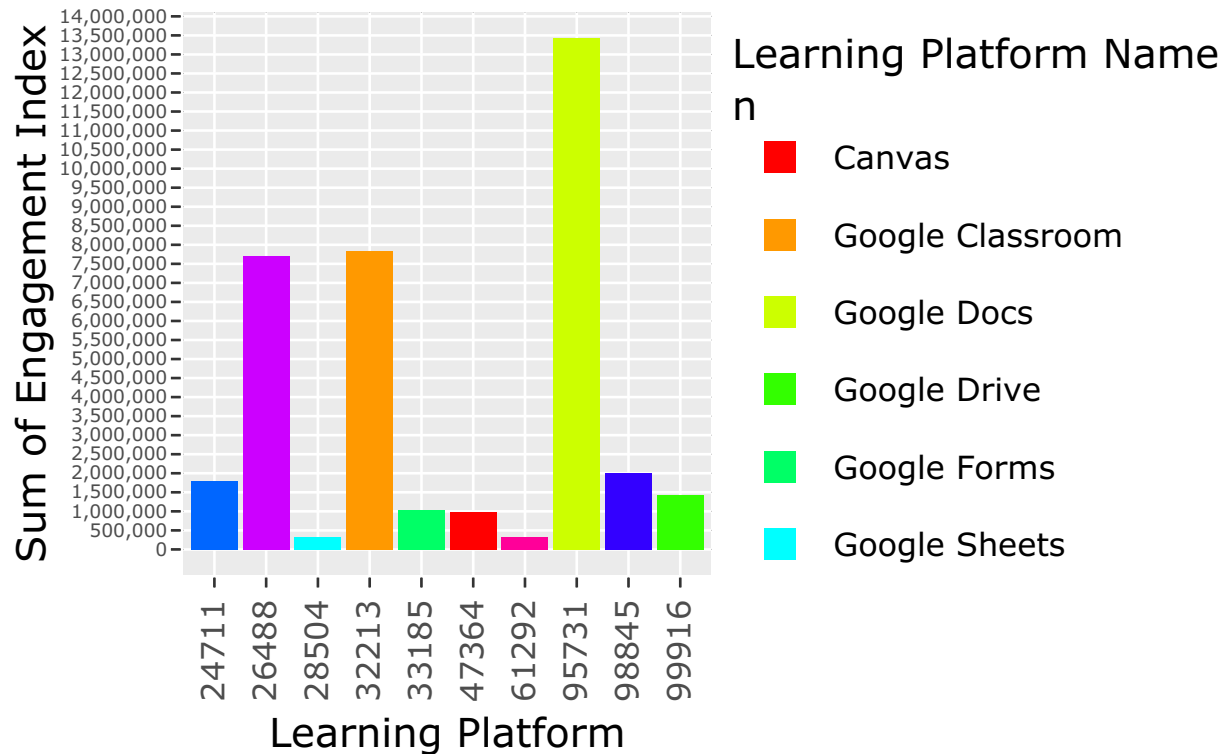
Percentage Access Growth in 2020



- I am now focusing on the top 10 learning platforms, and plotting their respective engagement data.

```
graph8 = ggplot(top_10_product, aes(x = lp_id, fill = product_name, y = total_engagement)) +
  geom_bar(stat = "sum", na.rm = FALSE) +
  labs(x = "Learning Platform", y = "Sum of Engagement Index") +
  ggtitle("Top 10 Learning Platform Based on 5 District Samples") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  scale_fill_manual(values = rainbow(length(unique(top_10_product$product_name)))) +
  labs(fill = "Learning Platform Name") +
  scale_y_continuous(labels = scales::comma, breaks = seq(0, 30000000, by = 500000)) +
  theme(axis.text.y = element_text(size = 5)) # Adjust font size for y-axis labels
# Convert ggplot to plotly
ggplotly(graph8)
```

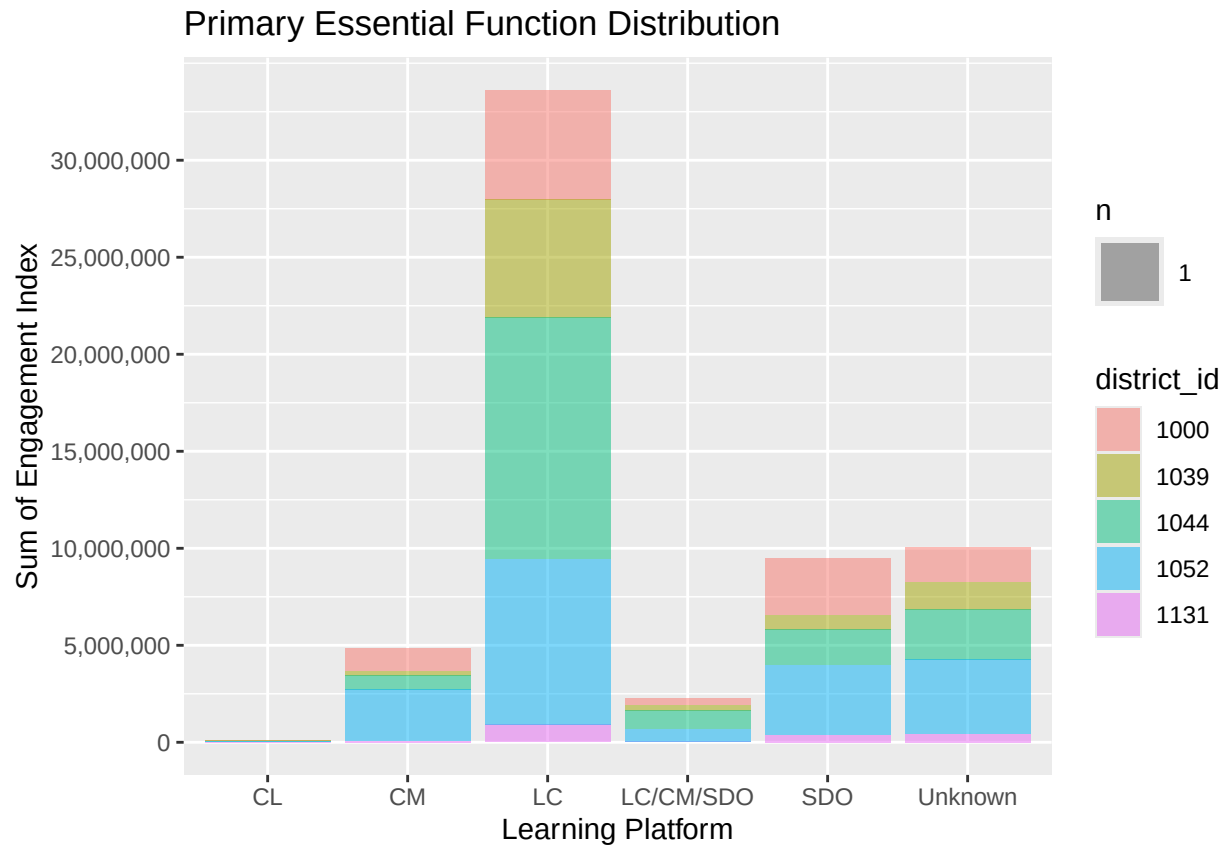
Top 10 Learning Platform Based on 5 District



3. How does student engagement with different types of education technology change over the course of the pandemic?

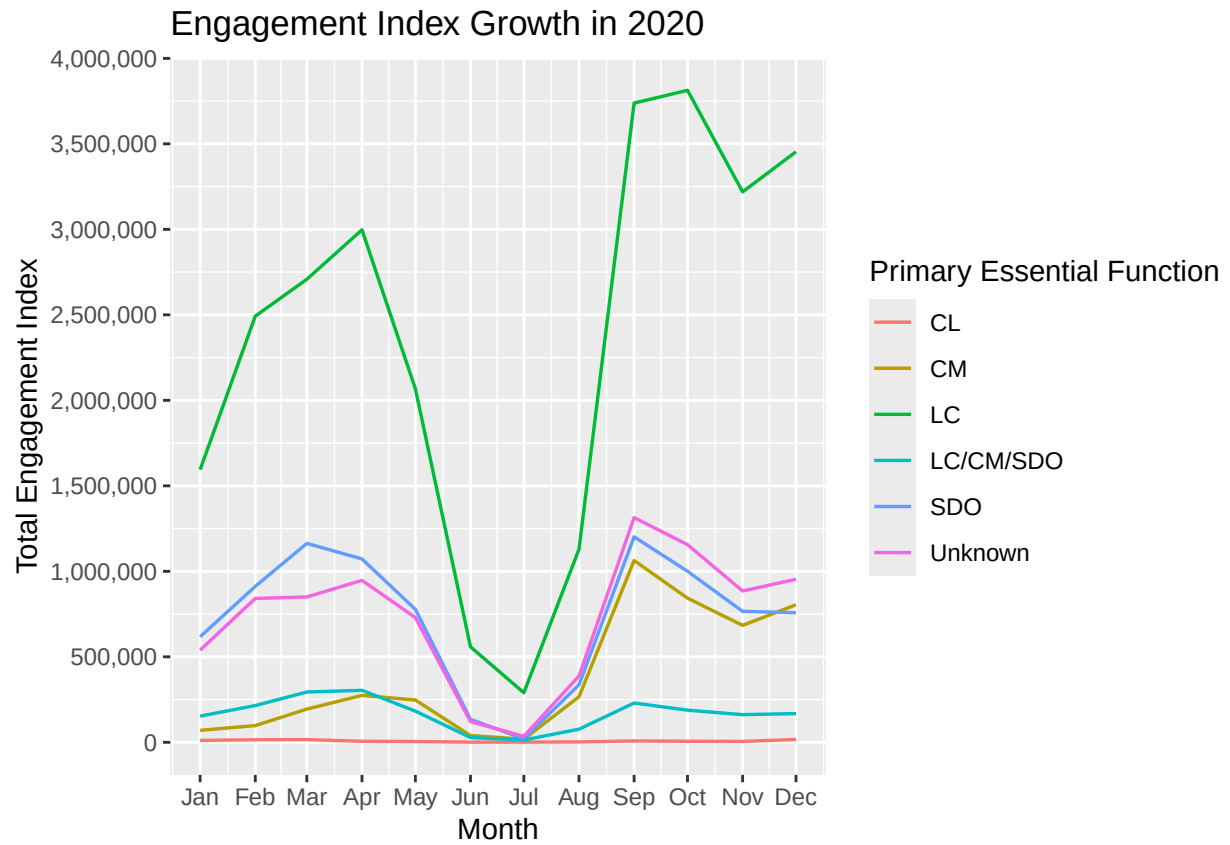
- I have taken the primary essential function data (LC = Learning & Curriculum, CM = Classroom Management, and SDO = School & District Operations) to further analyze the data.

```
ggplot(data3, aes(x = pef_cat, y = sum_engagement, fill=district_id)) +
  geom_bar(stat='sum',alpha=0.5) +
  labs(x = "Learning Platform", y = "Sum of Engagement Index") +
  ggtitle("Primary Essential Function Distribution") +
  scale_y_continuous(labels = scales::comma,breaks = seq(0, 30000000, by = 5000000))
```



- A line graph is created to track the changes in how engaged users are with different kinds of learning environments monthly. This was done in order to obtain insight into how the COVID-19 epidemic affected the development of learning systems.

```
ggplot(data4, aes(x = months, y = sum_engagement, color = pef_cat)) +
  geom_line(linewidth=0.6) +
  xlab("Month") +
  ylab("Total Engagement Index") +
  ggtitle("Engagement Index Growth in 2020") +
  xlim(as.Date("2020-01-01"), as.Date("2020-12-31")) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  scale_y_continuous(labels = scales::comma, breaks = seq(0, 30000000, by = 500000)) +
  labs(color = "Primary Essential Function")
```



3. Findings

Clearly explain your findings and provide a valid conclusion of your analysis. References may be used if required.(max 500 words)

Overview of the datasets and the data -The bar chart gives a comprehensive insight into the data volume that we have from all the datasets. We can see that District 1044 has provided the largest volume of data, while district 1131 has very low engagement data. -In essence, District 1131 lags behind considerably in Data sharing, whereas District 1044 has a strong commitment to data exchange. However, Districts 1000, 1039, and 1052 show a continuous and stable pattern of data contributions, making a consistent contribution to the engagement study.

Importance of Broadband Access -The scatterplot strongly suggests the positive correlation between engagement and digital connectivity. In other words, people are more likely to participate effectively in online learning and other digital activities when they have a strong digital connection, which includes dependable internet access and the appropriate gadgets. -People can participate in digital activities more easily when there is strong digital connectivity; weak or restricted connectivity can hinder them. This division emphasizes how important it is to have access to digital tools for meaningful interaction in today's digital world, even in a formal setting, as the scatterplot makes clear.

Engagement Index and Percentage Access Growth variables -The influence of the COVID-19 pandemic on online and distance learning in 2020 is depicted in both line graphs. It shows clear trends, with a discernible decline in the engagement index and percentage access (pct access) between June and August. Subsequent data, however, indicates that five districts show a recovery phase, as indicated by an upward trend in the engagement index and percentage access. -Notably, the effects of the post-pandemic period varied considerably throughout districts. Districts 1052 and 1000, for example, showed a notable rise in participation and access percentage as compared to the pre-COVID era. This suggests that the pandemic accelerated the adoption of technology and online learning in this district, pushing them to embrace digital tools and resources more extensively and rapidly than before. -On the other hand, district 1039 displayed a different pattern. While it did show signs of recovery, it was not able to fully regain the levels of engagement and percentage access seen before the COVID-19 pandemic. This indicates that district 1039 faced ongoing challenges or adjustments in the post-pandemic period that affected its online learning outcomes, showing that the impact of the pandemic varied across districts.

Analysing the learning platforms -We now focus on the usage of learning platforms in the top five districts, specifically the top 10 learning platforms, based on engagement data. We can see from the bar graph that 'Google Docs' has done tremendously, with the highest level of engagement amongst all the learning platforms. -One intriguing aspect about this data is that there is a variable, 'unknown', which shows a considerable volume of data, more than some of the top learning platforms.

Primary Essential Function - Based on the line and bar graph, we can see that the "Primary Essential Function" category mostly uses the "LC" (Learning & Curriculum) learning platform, followed by "SDO" (School & District Operations), and "CM" (Classroom Management) comes last. The chart shows that the "LC/CM/SDO" combination has a much lower total engagement score. This is because there are not many learning platforms that cover all the essential functions required by the "Primary Essential Function" category. LC also has the highest number of engagement because majority of the learning platforms available in 2020 are categorized as LC.